

EE045-4-3

Group Design Project

Student

Handbook

(Version 6 – December 2025)

School of Engineering

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1 Introduction to Group design project

1.1 Objectives:

The engineering group design project is a stand-alone course similar to other subjects in the engineering programme degree. In this module, you will form a team and complete a group design work project incorporating all aspects of design from identifying a specific problem to a fully functional sustainable prototype and presentation in both written and oral formats.

In the module, students will Identify a complex engineering problem and prepare preliminary work, Present project proposals through research and analysis evidenced by a critical thinking framework such as 5W1H, Propose solutions to the complex problem by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations, Evaluate and recommend a sustainable solution, by incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, with an inclusion of **at least ten United Nations-Sustainable Development Goals (UN-SDGs)**, Design a system, a component/device, or a computer program, Prepare design documentation and Review, Incorporate Engineering Economics, Engineering Management Tools and Techniques, Entrepreneurship and Environmental considerations, Design test and Validations, and Present complete design work effectively within multi-disciplinary settings as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings and through effective communication both orally and in written context by taking into account cultural, language, and learning differences such as Visual learners, Auditory learners, Kinesthetic learners, Social learners, Solo learners.

Thus, this module is to expose you to the process of engineering design management and practices through the appropriate use of skills and knowledge learned throughout the programme. You are expected to utilize what you have learned in the previous modules (fundamental principles), learn new skills or concepts and include complex engineering problem (refer to appendix C) and achieve a design system, components, or processes integrating the core areas that meets the desired needs within the realistic constraints and within the time limit provided. You are expected to demonstrate the complex problem solving and analytical skills with appropriate consideration for public health and safety, cultural, societal and environmental considerations.

1.2 Course & Programme Learning Outcomes:

Course Learning Outcomes Attained:

At the end of the Group Design Project module, you should be able to:

1. Review the existing methods for the complex engineering problem given by a critical thinking framework such as 5W1H. (C5, PLO4)
2. Develop solutions to implement proposed methodology for the given complex

- engineering problem by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations. (C6, PLO3)
3. Explain professional engineering practices for safety, health, social, cultural and legal responsibilities in developing solutions by incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, with an inclusion of at least ten United Nations-Sustainable Development Goals (UN-SDGs). (C5, PLO6)
 4. Review the sustainability and environmental considerations to the implemented solution by incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, with an inclusion of at least ten United Nations-Sustainable Development Goals (UN-SDGs). (C5, PLO7)
 5. Explain project management execution, economic decision making and entrepreneurial considerations for the implemented solution. (C5, PLO10)
 6. Demonstrate the ability to work effectively within multi-disciplinary settings as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings. (A3, PLO8)
 7. Defend the implemented solution through effective communication both orally and in written context by taking into account cultural, language, and learning differences such as Visual learners, Auditory learners, Kinesthetic learners, Social learners, Solo learners. (A5, PLO9)

Program Learning Outcomes Attained:

PLO3: Design creative solutions for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, whole-life cost, net-zero carbon, resource efficiency, and cultural, societal, and environmental factors.

PLO4: Conduct investigations of complex engineering problems using research-based knowledge, design of experiments, data analysis and interpretation, and synthesis of information to provide valid conclusions.

PLO6: Analyse and evaluate the impacts of engineering solutions on society, the economy, sustainability, health and safety, legal frameworks, and the environment, as represented by the 17 UN Sustainable Development Goals, with strong social awareness.

PLO7: Apply ethical principles and commit to professional ethics and norms of engineering practice, comply with relevant laws, and demonstrate understanding of diversity, equity, and inclusion in engineering contexts.

PLO8: Function effectively as an individual, and as a member or leader in diverse and inclusive teams, including multidisciplinary, face-to-face, remote, and distributed working environments.

PLO9: Communicate effectively and inclusively on complex engineering activities with the engineering community and society at large, through effective reports, documentation, and presentations that consider cultural, language, and learning differences.

PLO10: Apply knowledge and understanding of engineering management principles and economic decision-making to one's own work and to manage projects in multidisciplinary environments.

In this module, each PLO is directly and explicitly mapped to the CLOs as shown in **Appendices E & F**. Percentage of individual PLOs measured in this module is given below:

CLO	Learning Outcome	PO Assessed	Taxonomy Assessed	Component assessed	Percentage of Marks
CLO1	Review the existing methods for the given complex engineering problem evidenced by a critical thinking framework such as 5W1H.	PLO4	C5	Individual System – Literature Review	5%
CLO2	Develop solutions to implement proposed methodology for the given complex engineering problem by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations.	PLO3	C6	Individual System implemented and Integrated system and Enhancement	20%
CLO3	Explain professional engineering practices for safety, health, social, cultural and legal responsibilities in developing solutions by incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, with an inclusion of at least ten United Nations-Sustainable Development Goals (UN-SDGs).	PLO6	C5	Professional Engineering Practice	5%
CLO4	Review the sustainability and environmental considerations to the implemented solution by incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon,	PLO7	C5	Sustainability & Environmental considerations	10%

	with an inclusion of at least ten United Nations-Sustainable Development Goals (UN-SDGs).				
CLO5	Explain project management execution, economic decision making and entrepreneurial considerations for the implemented solution.	PLO10	C5	Project Management, finance and Entrepreneurship	10%
CLO6	Demonstrate the ability to work effectively within multi-disciplinary settings as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings.	PLO8	A3	Proposal, Mid-point and Final Individual Presentation	40%
CLO7	Defend the implemented solution through effective communication both orally and in written context by taking into account cultural, language, and learning differences such as Visual learners, Auditory learners, Kinesthetic learners, Social learners, Solo learners.	PLO9	A5	Poster Presentation	10%

Thus, **the group design project work is** weighted at 100%, with 35% of the total contributed by an individual component and 65% of group component, with 50% for report and 50% for presentation.

1.3 GDP process flow

The Module lecturer introduces the module to the student during week 1 and hands out the assignment during the same week. During week 2, students are exposed to sustainability, environmental considerations, professional engineering practices, entrepreneurship, complex engineering activities, complex problem solving and GDP proposal presentation.

During week 3-4, formation and registration of the group to module lecturer shall be done. Each group shall present their proposal with current problem clearly stated, along with aim & objectives. Shall propose the methodology with new skills/ techniques / creativity /enhancement embedded, demonstrating sustainability along with the timeline to complete the work. This proposal presentation shall be evaluated by the Module lecturer along with two other experts of different programme called as GDP team, contributing to

10% group work mapped to CLO6-PLO8.

Upon approval, the group shall progress independently and the group leader shall conduct group meetings appropriately and monitor the progress of the group members according to the proposed timeline, for the next three weeks.

During week 8, Mid-point group presentation shall be conducted where the group leader presents the aim & objectives of the GDP, followed by the group members presenting their work done and finally the group leader concludes with the timeline to complete the work left out. This is again evaluated by the same GDP team, contributing to 5% group work and 10% individual work mapped to CLO6-PLO8.

Again, the group shall progress independently and the group leader shall conduct group meetings appropriately and monitor the progress of the group members according to the proposed timeline, for the next three weeks.

The group shall submit their GDP report during week 12, contributing 35% group work and 15% individual work. During week 13, final individual presentation is done, where the group leader presents the aim & objectives of the GDP, followed by the group members presenting their final work done and lastly, the group leader concludes on the aim & objectives. This is again evaluated by the same GDP team, contributing to 5% group work and 10% individual work mapped to CLO6-PLO8.

Lastly, poster presentation is done during week 14, where the group work is evaluated by the same GDP team along with/or an invited expert from industry, contributing to 10% group work mapped to CLO7-PLO9.

1.4 Group Design Project Implementation:

The Module Lecturer explains the entire process of the group design project.

Step 1: Group Formation

First a group has to be formed with two or three members and maximum of four members, involving a balanced mixture of students from different programme (CE/EE/ME/TE) depending upon the strength of the class. The Group must have a group name and the group must elect a group leader. Each team member of the group has to involve and develop the assigned task by the group leader

Step 2: Proposal Presentation & Approval

Once the group has been formed, the group has to present the proposal for approval.

The presentation must include the following:

- Must introduce the group with the group leader followed by the group members
- Current problem statement with reference
- Aim and objectives
- Justification for the research to be done
- Literature review – minimum of three current research papers with critical thinking framework such as 5W1H
- Proposed methodology with the help of a block diagram or flow chart, by incorporating

public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations.

- g. Justification on the enhancement or creativity or innovativeness included
- h. Justification on how the group project is sustainable, at least ten United Nations-Sustainable Development Goals (UN-SDGs) to be mapped
- i. Justification on how complex engineering problem is achieved
- j. Gantt chart showing the group component and the individual components allocated. The individual work allocated must be equally divided among the group members.
- k. References

The module lecturer approves based on the criteria of g to j being satisfied. If not, the group will be asked to update accordingly and present again until the proposal is approved. The proposal approval must be done within the first six weeks.

Step 3: Individual Presentation

The individual presentation must cover the following:

- a. Introduction
- b. Literature review
- c. Methodology
- d. Investigation on tools/ techniques selected
- e. Concept design derived from fundamental engineering principles
- f. System implementation – constructional details, working principle, results, and testing (data collection & data analysis)
- g. Summary

Step 4: Group Presentation

All the individual work carried out by team members must be integrated and presented.

The group presentation must cover the following:

- a. Introduction
- b. Integrated system & Enhancement– overall block diagram & working principle, testing and results
- c. Summary
- d. Professional Engineering Practices
- e. Sustainability and Environmental
- f. Project Management, Entrepreneurship and Finance – Gantt chart, marketing strategy and cost analysis
- g. Discussion – result analysis, sources of errors and troubleshooting methods employed
- h. Conclusion – relating to the objectives, limitations and future work

1.5 Group Design Project Assessment:

The group design project work is weighted at 100%, with 35% of the total contributed by an individual component and 65% of group component, with 50% for report and 50% for presentation.

Assessment Breakdown

1 INDIVIDUAL WORK (35%)

1.1 Review on the existing methods (5%) CLO1-PLO4-Cognitive C5

- a) **Review** on the past related work done by other researchers.
- b) **Justify** on the components to be used with proper comparison or investigation.
- c) **Reflection of PL04 evidenced by a critical thinking framework such as 5W1H.**

Individual component (10%) – either 1.2 or 1.3 or 1.4-CLO2-PLO3-Cognitive C6

1.2 Electronic / Mechatronic Circuit design development

- a) One of the team members must develop conceptual electronic / mechatronic circuit design of the system.
- b) **Compose** the circuit printed on PCB is working.
- c) **Develop** appropriate programs for the control application of the project.
- d) **Compose** on the individual test results obtained (data analysis), problems faced and troubleshooting methods employed with limitations.
- e) **Reflection of PL03 evidenced by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations.**

1.3 Transceiver design development

- f) One of the team members must design the transceiver of the system.
- a) **Compose** the circuit working.
- b) **Develop** appropriate programs for the same.
- c) **Compose** on the individual test results obtained (data analysis), problems faced and troubleshooting methods employed with limitations.
- d) **Reflection of PL03 evidenced by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations.**

1.4 Graphical User Interface (GUI) platform development

- a) One of the team members must be able to develop the GUI.
- b) **Develop** the GUI using appropriate software.
- c) **Compose** the GUI developed

- d) **Compose** on the individual test results obtained (data analysis), problems faced and troubleshooting methods employed with limitations.
- e) **Reflection of PL03 evidenced by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations.**

1.5 **Demonstration of the distinct individual component working (20%)-CLO6-PLO8-Affective A3**

- a) **Demonstrate** the individual progress work done during the Mid-point Group presentation at week 8-9 and the final individual presentation at week 15.
- b) Each team member must be able to answer questions related to the individual component during the project presentation.
- c) **Reflection of PLO8 in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings.**
- d) **Reflection of PLO8 in individual Effectiveness and Peer Review.**

2 **GROUP WORK (65%)**

2.1 **Introduction**

- a) **Explain** the project background with clear problem statement and justification for the project.
- b) State the aim and objectives of the project

2.2 **Designing the Complete system (10%)-CLO2-PLO3-Cognitive C6**

- a) Develop the intended system and demonstrate the working principle.
- b) **Compose** the function and working of component used in the system.
- c) **Design** the circuit using adequate PCB software and fabricate.
- d) **Compose** on the enhancement done.
- e) **Reflection of PL03 evidenced by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations.**

2.3 **Professional Engineering Practices (5%)-CLO3-PLO6-Cognitive C5**

- a) **Explain** on how the Professional engineering practices are adhered in developing your proposed method solutions
- b) **Explain** on the standards, rules, laws for safety, health, social, cultural, and legal responsibilities considerations taken into account.
- c) **Reflection of PLO6 by incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero**

carbon, with an inclusion of atleast ten United Nations-Sustainable Development Goals (UN-SDGs).

2.4 Sustainability and Environmental (10%)-CLO4-PLO7-Cognitive C5

- a) **Explain** on how sustainability is achieved in your proposed method.
- b) **Explain** on the environmental considerations taken into account.
- c) **Reflection of PLO7 by incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, with an inclusion of atleast ten United Nations-Sustainable Development Goals (UN-SDGs).**

2.5 Project Management, Finance and Entrepreneurship (10%)-CLO5-PLO10-Cognitive C5

- a) **Explain** on how the Group design project is managed with the help of a Gantt chart.
- b) **Explain** the cost involved with the help of a tabulated cost incurred.
- c) **Explain** on the targeted market and considerable number of trends on marketing.

2.6 Discussion and Conclusion

- a) Organise the test results obtained (data analysis), problems faced and troubleshooting methods employed, limitations and future recommendations.
- b) Conclude on the aim and objectives.

2.7 Group project proposal presentation (5%)-CLO6-PLO8- Affective A3

- a) **Justify** the proposed method with all the aim and objectives of the project proposed in order to solve the current complex engineering problem during the Group project proposal presentation at week 3-4, based on the independent research done.
- b) Students must be able to answer questions regarding the project proposed for implementation.
- c) **Reflection of PLO8 - within multi-disciplinary settings as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings.**

2.8 Overall project working demonstration during Poster Presentation (10%) – CLO7-PLO9-Affective A5

- a) **Defend** the implemented solution through oral communication, in achieving all the objectives of the project during the poster presentation at week 14.

- b) Students must be able to answer hardware and software related question regarding the project during the demonstration.
- c) **Reflection of PLO9 by taking into account cultural, language, and learning differences such as Visual learners, Auditory learners, Kinesthetic learners, Social learners, Solo learners**

2.9 Group Project updates (15%)-CLO6-PLO8-Affective A3

- a) **Demonstrate** the Group progress work done with the help of a Gantt chart during the Mid-point Group presentation at week 8 and the final individual presentation at week 13.
- b) Workflow and timelines must be met.
- c) **Reflection of PLO8 - within multi-disciplinary settings as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings.**

1.6 Report Writing Format

- a) Introduction (Group Work)
Discuss the background, problem statement, justification, aim and objectives, and literature review.
- b) Electronic / Mechatronic Circuit working (Individual component)
 - Review on the past related work done by other researchers.
 - Compare on the tools and techniques selected with appropriate justification.
 - Explain how the circuit works and show the circuit diagram and PCB development.
- c) Graphical User Interface (GUI) programming (Individual component)
Use flow chart to explain the program flow.
- d) Transceiver (Individual component)
Explain the overall transceiver circuit and show any analytical derivation/ calculations for the design.
- e) Integrated system and Enhancement (Group Work)
- f) Professional Engineering Practices (Group Work)
- g) Sustainability and Environmental (Group Work)
- h) Project Management, Finance and Entrepreneurship (Group Work)
- i) Individual Reflection (Individual Work)

- j) Meeting Modes (Group Work)
- k) Discussion and Conclusion (Group Work)
- l) References (Group Work)

1.7 Group Design Project Submission Timeline

- 1) **Failing to attend any or both presentation on the stated time will result in failing the presentation component and the marks for report will be penalised as the student's work cannot be validated.**

Week	Event	Activity	Assessment Marks %
1	Module Introduction & Group assignment is handed out	Student is introduced to the module by the module lecturer	
2	Sustainability & Report writing	Sustainability is lectured & briefing done on report writing by the module lecturer	
3-4	Group registration & Proposal Presentation & Approval	GDP team is invited to evaluate the proposal along with the module lecturer (2 members + ML)	5% (Group Component)
5-7	Independent work		
8	Mid-point Group presentation	Presentation done as a group, presenting the individual work done. GDP team is invited to evaluate the proposal along with the module lecturer (2 members + ML)	5% (Individual Component) + 5% (Group Component)
9-10	Independent work		
11	Report Submission Briefing	Briefing done by the Module lecturer	
13	Poster Preparation & Final Presentation Briefing	Briefing done by the Module lecturer Final Individual Presentation (2 members + ML)	5% (Individual Component) + 5% (Group Component) + 5% (Individual Reflection)
14	Poster Presentation	Poster evaluated by the IAP/Industry Panel along with the GDP team (1+2 members)	10% (Group Component) + 5% (Meeting Modes-Group) + 5% (peer Review-Individual)
14	Report Submission		50% (Report Submitted Group – 35%; Individual – 15%)
	TOTAL (Group work – 65% & Individual work – 35% ; Report – 50% & Presentation – 50%)		100%

Group Design Project Report Writing Format

2.1 Group Design Project Report Writing Format (Only one report to be submitted by each group)

The project may be substantial and may take a variety of forms but the report should not exceed **15,000 – 20,000** words. DO NOT try to produce the biggest report possible. Quality is more important than quantity.

There are standards expected for the presentation of a report. You need not worry about these at the start of the group project. Full details are supplied separately in Appendix A.

Typically the format of the report should contain the following sections:-

Front cover Page

- Use for both hard cover and inside report
- follow format as attached in the Appendix A

Declaration of Originality

- follow format as attached in the Appendix A

Acknowledgement

- follow format as attached in the Appendix A

Abstract

- outline the problem
- give summary of the results
- not more than ½ page in length

Table of Contents

- include page numbers

List of Figures

- follow format as attached in the Appendix A

List of Tables

- follow format as attached in the Appendix A

List of Symbols, Abbreviations and Nomenclature

- follow format as attached in the Appendix A

List of Appendices

- follow format as attached in the Appendix A
- Organization of the rest of the chapters

Introduction to the study

- a. Introduction
- b. Research problem
- c. Aim and objectives
- d. Justification to the research
- e. Organization of the rest of the chapters
- f. Summary

Individual chapters

- a. Introduction
- b. Literature review
- c. Investigation on materials/ components selection
- d. Methodology
- e. Concept design derived from fundamental engineering principles
- f. System implementation – constructional details, working principle, results, and testing (data collection & data analysis)
- g. Summary

Integrated System and Enhancement

- a. Introduction
- b. Integrated system and Enhancement – overall block diagram & working principle, testing and results
- c. Summary

Professional Engineering Practices

Sustainability and Environmental

Project Management, Finance and Entrepreneurship

- a. Gantt chart,
- b. Cost analysis, and
- c. Marketing

Self-Review

Meeting Modes

Discussion and Conclusion

- a. Discussion must include result analysis, sources of errors and troubleshooting methods employed and
- b. Conclusion must be relating to the objectives, limitations and future work

REFERENCES

References/Bibliography

- follows format as attached in the Appendix B

APPENDICES

Depending on Project

- Appendix A : Material selection (optional)
- Appendix B : Program code
- Appendix C : Data sheet
- Appendix D : Peer Review Form

2.2 Group Design Project Report Specifications

2.2.1 Page Dimension and Binding Specification:

The dimension of the project report should be in A4 size. Submission should be made online as per instructions from your Module Lecturer. All the pages shall be numbered consecutively through the main text including photographs and/or diagrams included as whole pages.

2.2.2 Abstract:

It should be to half a page synopsis of the project typed double line spacing, Font Style Times New Roman and Font Size 12.

2.2.3 Table of Contents:

The table of contents should list all the topics following it as well as any topic which precedes it. The title page will not find a place among the items listed in the table of contents. **The page numbers starting from the front page must be counted with lower case Roman letters until the list of appendices, centered at the bottom of the page in Times New Roman 12 font style. Page numbers must be displayed from the second page onwards.** One and a half spacing should be adopted for typing the matter under this head. A specimen copy of the Table of Contents of the project report is given in **Appendix A**.

2.2.4 List of Figures:

The list should be exactly the same captions as they appear below the figures in the text. One and a half spacing should be adopted for typing the matter under this head. The captions should be placed below the figure.

2.2.5 List of Tables:

The list should be exactly the same captions as they appear above the tables in the text. One and a half spacing should be adopted for typing the matter under this head. The captions should be placed above the table.

2.2.6 List of Symbols, Abbreviations and Nomenclature:

One and a half spacing should be adopted or typing the matter under this head. Standard symbols, abbreviations etc. should be used.

2.2.7 List of Appendices:

One and a half spacing should be adopted or typing the matter under this head. Standard symbols, abbreviations etc. should be used.

2.2.8 Chapters

The chapters are broadly divided as specified in section 2.1. The general text should be typed using the Font Style “Times New Roman, 12” with 1.5 line spacing. The page numbering must start from chapter 1 onwards, centered with the bottom of the page in Times New Roman 12 font style, with decimal number system. Equation numbering must be justified with curved brackets as per the chapters as shown below.

Example:

$$v_{FM}(t) = V_c \cos(2\pi f_c t + m_f \sin(2\pi f_m t)) \dots\dots\dots(3.1)$$

$$m_{FM-NB}(t) = A \cos(2\pi f_0 t + \phi(t)) \dots\dots\dots(3.2)$$

2.2.9 References

The listing of references should be typed 4 spaces below the heading “REFERENCES” in alphabetical order in 1.5 line spacing left – justified. The reference material should follow APA referencing style as in **Appendix B**.

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Appendix A

Specimen

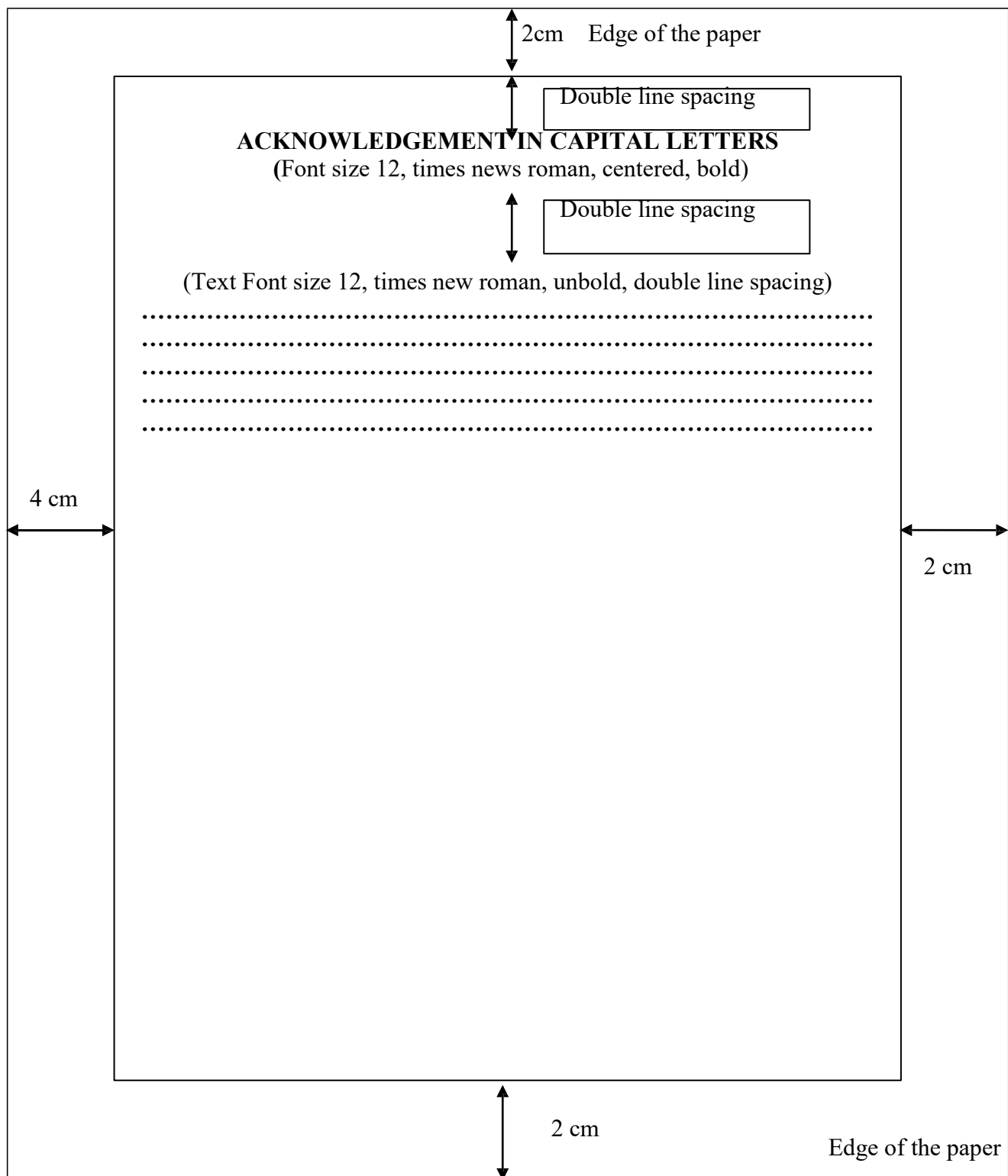
Specimen of typical cover and front page for Group Design project report:

	2 cm	Edge of the paper
		
	PROJECT TITLE IN CAPITAL LETTERS, FONT SIZE 12, TIMES NEWS ROMAN, CENTERED, BOLD	
	STUDENT NAME	
	STUDENT NUMBER	
	INTAKE	
	PROGRAMME	
	SUPERVISOR NAME	
	A Group project	
	ASIA PACIFIC UNIVERSITY OF TECHNOLOGY & INNOVATION	
	SCHOOL OF ENGINEERING	
	MONTH , YEAR	
4 cm	2 cm	Edge of the paper

Declaration of Originality:

Edge of the paper	
↕ 2cm ↕	
↕ 2 line spacing ↕	
DECLARATION OF ORIGINALITY IN CAPITAL LETTERS, (Font size 12, times news roman, centered, bold)	
↕ 2 line spacing ↕	
We hereby declare that this group project report is based on our original work except for citations and references which have been duly acknowledged. We also declare that we have not been previously and concurrently submitted for any other degree of award at APU or other institution. (Text Font size 12, times new roman, unbold, 1.5 line spacing)	
↕ 2 line spacing ↕	↔ 4 cm ↔
Signature: _____ Name of the student1: _____ TP No: _____ Date: _____	
↕ 2 cm ↕	
Edge of the paper	

Acknowledgement format:



Group Design Project Abstract format:

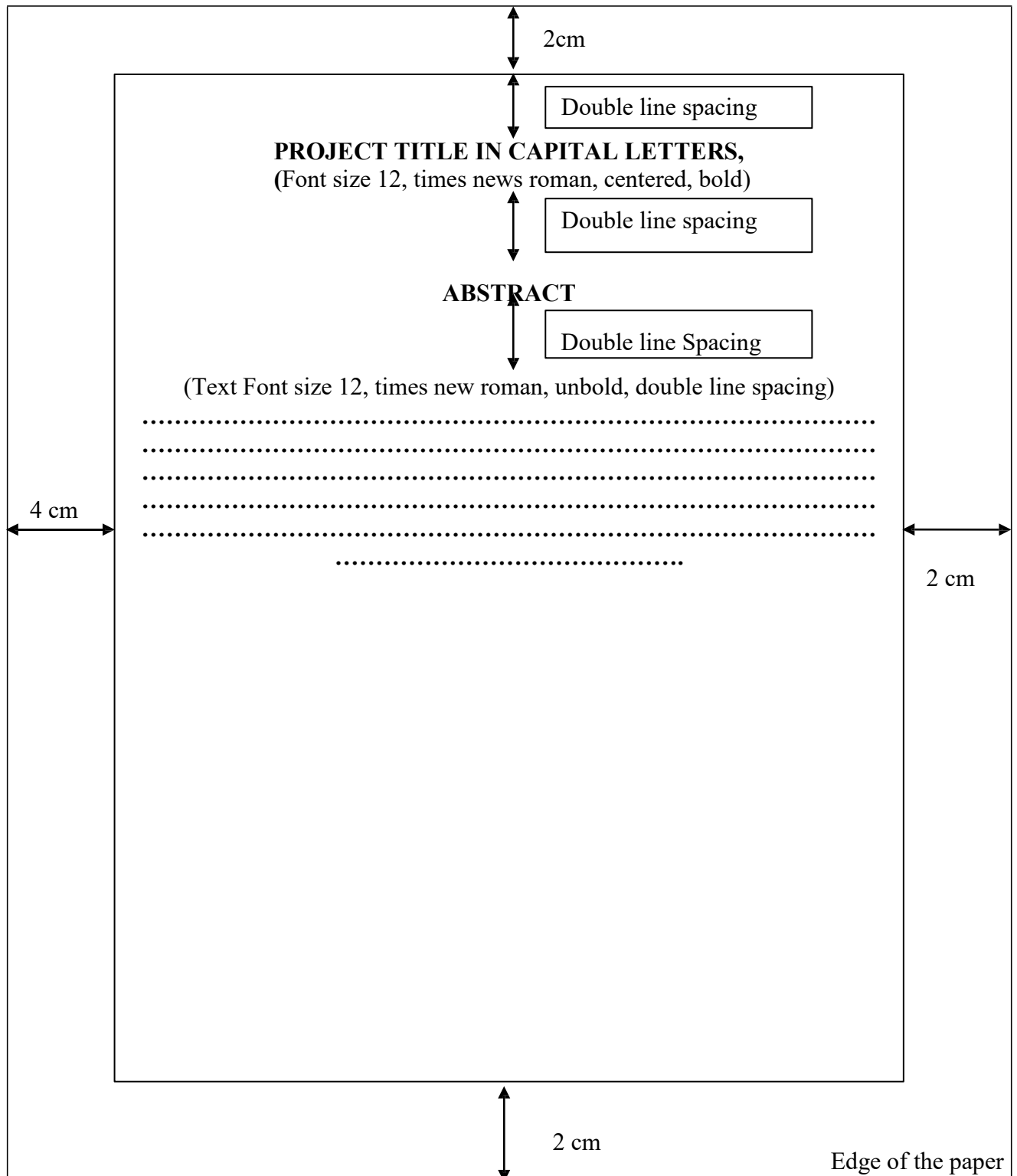


Table of Contents:

Double line spacing TABLE OF CONTENTS IN CAPITAL LETTERS, (Font size 12, times news roman, centered, bold) Double line spacing		
CHAPTER NO.	TITLE	PAGE NO.
0.5"		Double line Spacing
ABSTRACT		v
TABLE OF CONTENTS		vi
LIST OF FIGURES		vii
LIST OF TABLES		viii
LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE		x
LIST OF APPENDICES		xi
0.25"		Double line spacing
1. INTRODUCTION		1
0.25"	1.1 Introduction	1
	1.2 Research Problem	2
	1.3 Aim and Objectives	3
	1.4 Justification to this research	4
	1.5 Organization of the rest of the chapters	5
	1.6 Summary	6
0.25"		

		2cm	Edge of the paper
		Double line spacing	
CHAPTER NO.	TITLE	PAGE NO.	
		Double line Spacing	
2. CIRCUIT DESIGN		15	
2.1 Introduction		15	
2.2 Literature review		16	
2.2.1 Review based on sensors		16	
2.2.2 Review based on vision systems		20	
2.3 Methodology		25	
3. GRAPHICAL USER INTERFACE		20	2 cm
		2 cm	Edge of the paper

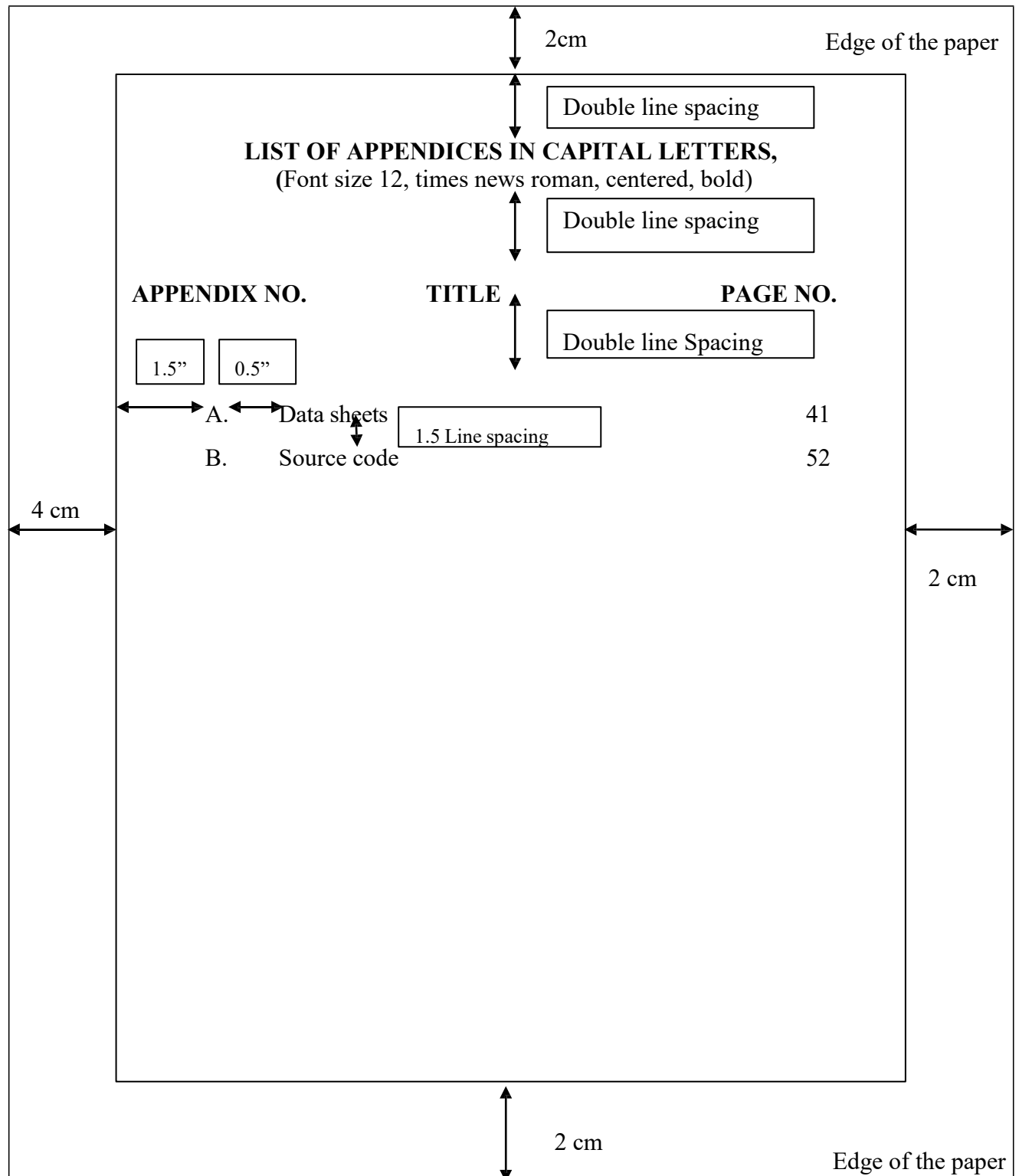
List of Figures:

LIST OF FIGURES IN CAPITAL LETTERS, (Font size 12, times news roman, centered, bold)			Double line spacing
			Double line spacing
			Double line Spacing
FIGURE NO.	TITLE	PAGE NO.	
0.5"	0.5"		
1.1	Block diagram	11	
1.2	System diagra	12	1.5 line spacing
1.3	Sensor	22	
2.1	Turbine	24	
3.1	Proposed block diagram	35	

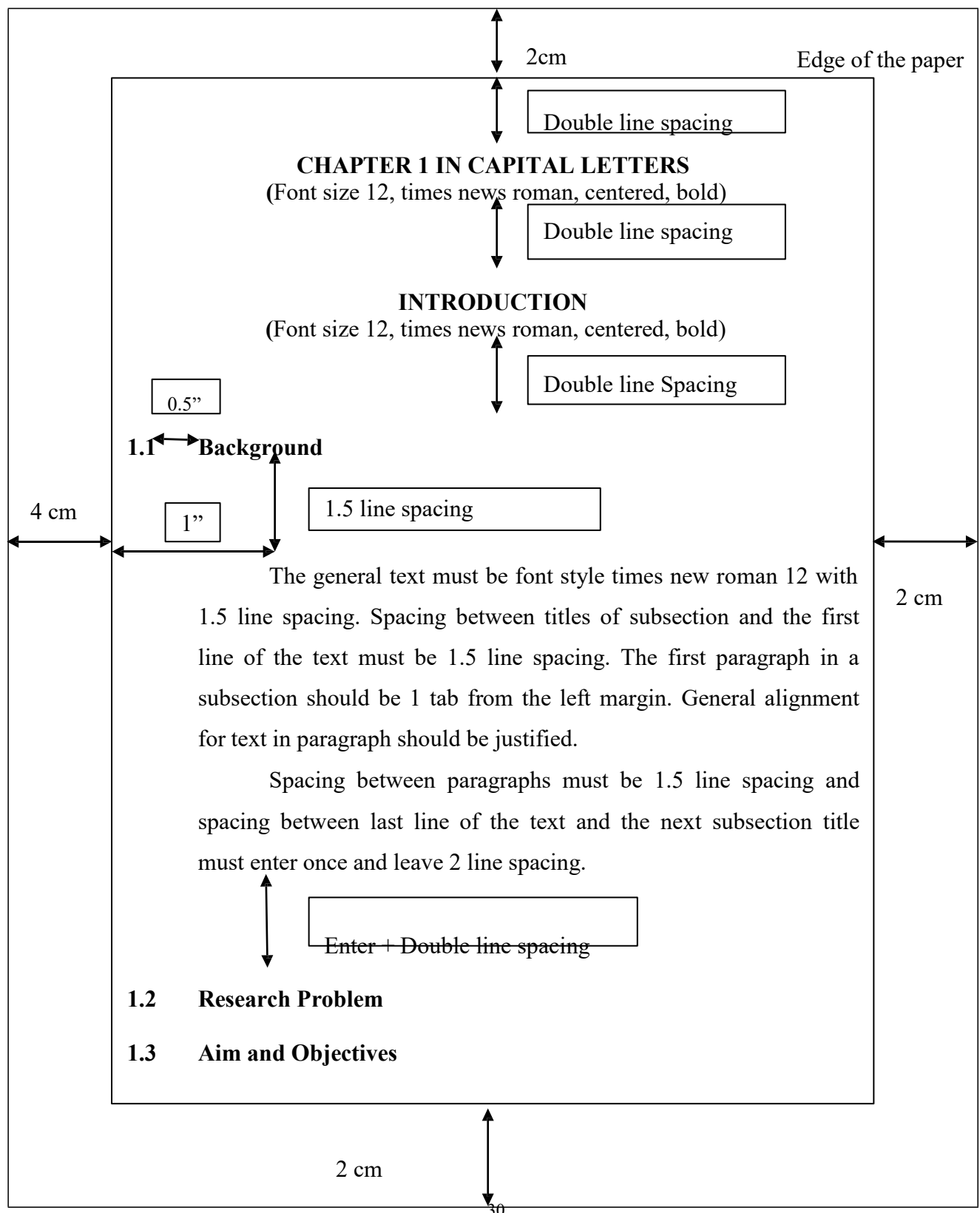
List of Tables:

			2cm	Edge of the paper
			Double line spacing	
LSIT OF TABLES IN CAPITAL LETTERS, (Font size 12, times news roman, centered, bold)			Double line spacing	
			Double line Spacing	
TABLE NO.	TITLE	PAGE NO.		
0.5"	0.5"			
1.4	Block diagram	11	1.5 Line Spacing	
1.5	System diagram	12		
1.6	Sensor	22		
2.2	Turbine	24		
3.2	Proposed block diagram	35		2 cm
			2 cm	Edge of the paper

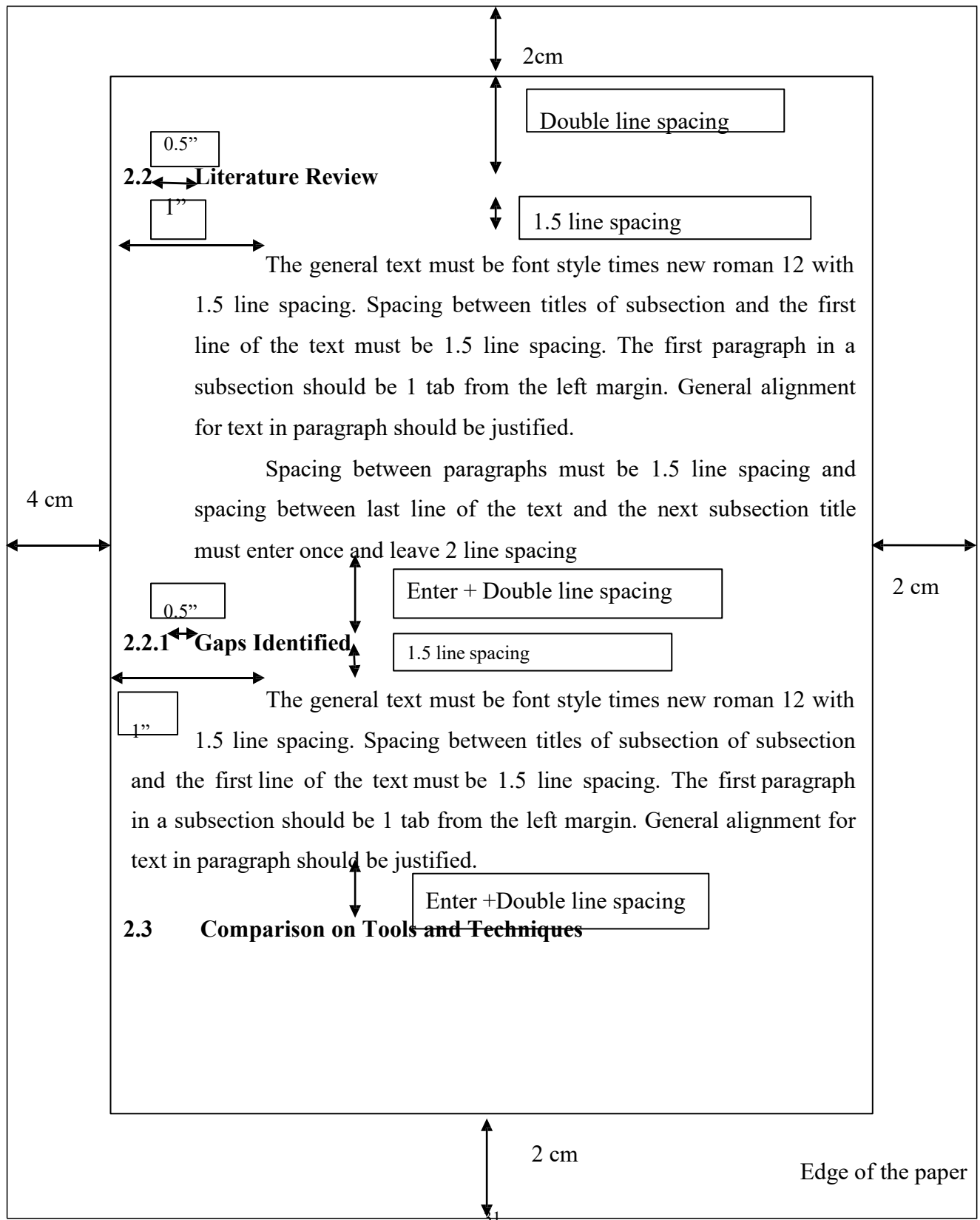
List of Appendices:



CHAPTERS:



CHAPTERS:



Appendix B

Citation and References

How to Cite References

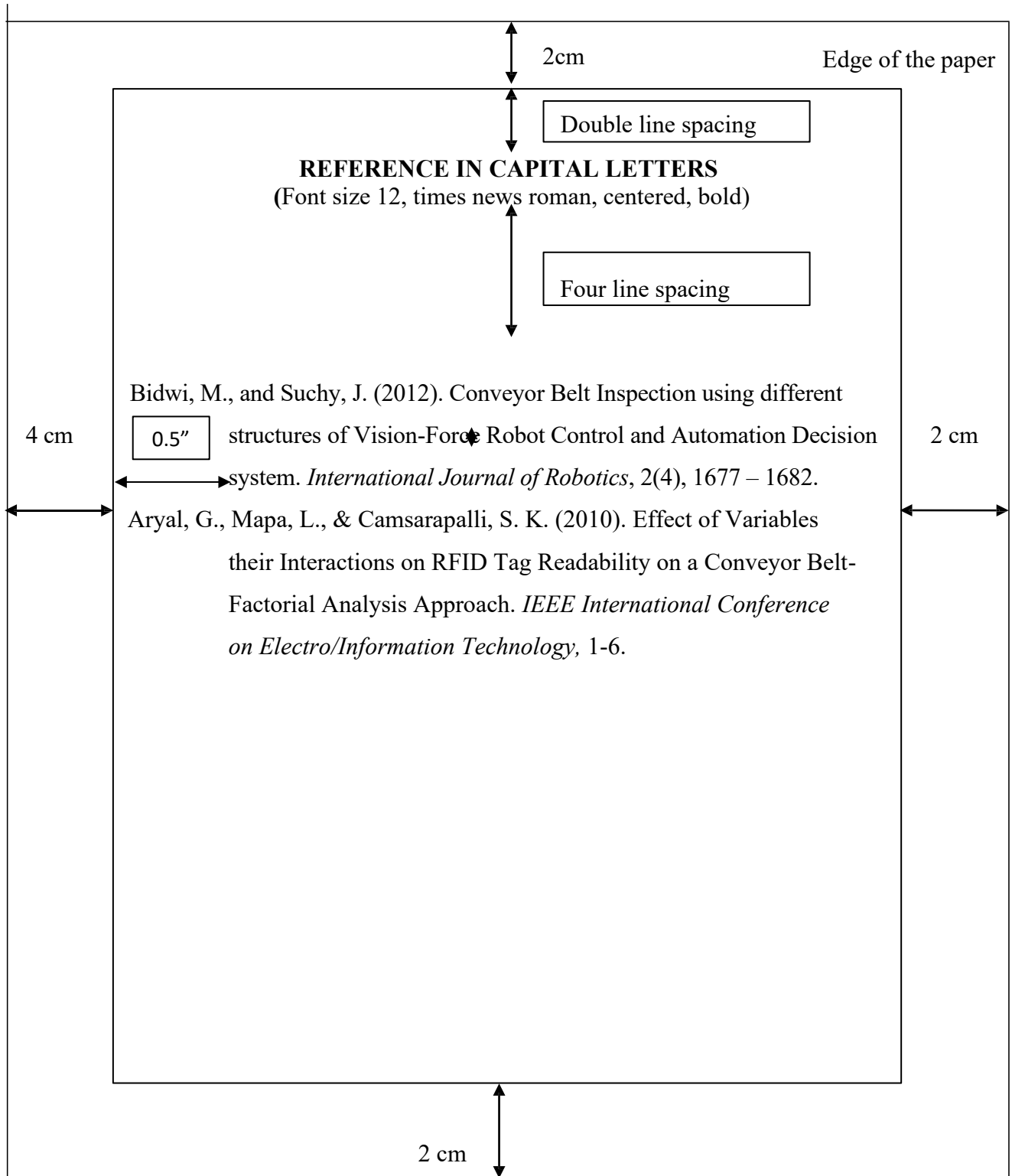
When you are writing your assignment or essay it is essential that you provide detailed and precise information on all the sources you have consulted (references). You may have used books, journal articles, newspapers, TV programmes, videos, the internet, government papers, statistics, etc. Every time you use quotations, or draw upon facts and arguments you must acknowledge your sources. This protects you from accusations of plagiarism (stealing other people's ideas and statements and passing them off as your own).

Citing your references also enables the reader to identify and trace the works that you used, and shows the authority on which you base your statements, demonstrates how well acquainted you are with the subject, and is a starting point for anyone else wanting to find out about the subject.

APA Referencing System

A number of methods exist for citing references. The Institution uses the APA Referencing Style, where APA stands for American Psychological Association.

REFERENCES SPECIMEN:



List of References:

All references used in carrying out the project and in producing the thesis must be listed in alphabetical order following the below given style and format.

Books:

Single author:

Format:

Family/Surname, Initials. (Year of Publication – in brackets). *Book Title* – in italics. Publisher.

Bibliography example:

Neville, C. (2010). *The Complete Guide to Referencing and Avoiding Plagiarism*. Open University Press.

In-text examples:

(Neville, 2010)

Neville (2010) commented that

“Direct quotations are placed in double quotations marks” (Author’s Surname, year of Publication, p. – followed by page numbers – in brackets)

Two authors:

Format:

Family/s Surname, Initials., & Family/Surname, Initials. (Year of Publication – in brackets). *Book Title* – in italics. Publisher.

Bibliography example:

Middleton, V. T. C., & Hawkins, R. (1998). *Sustainable Tourism: A Marketing Perspective*. Oxford: Butterworth- Heinemann.

In-text examples:

(Middleton & Hawkins, 1998)

As stated by Neville Middleton and Hawkins (1998)....

“Direct quotations are placed in double quotations marks” (First Author’s Surname & Second Author’s Surname, year of Publication, p. – followed by page numbers – in brackets)

Three or more authors:

Format:

Family/s Surname, Initials., Family/s Surname, Initials., & Family/s Surname, Initials. (Year of Publication – in brackets). *Book Title* – in italics. Publisher.

Bibliography example:

Bradbury, I., Boyle, J. & Morse, A. (2002). *The Scientific Principles for Physical Geographers*. Prentice Hall.

In-text examples:

(Bradbury et al., 2002)

As noted by Bradbury et al. (2002)....

“Direct quotations are placed in double quotations marks” (First Author’s Surname et al., year of Publication, p. – followed by page numbers – in brackets)

Journal article (printed journal article)

Printed article format:

Family/Surname, Initials. (Year of publication – in brackets). Title of article. *Title of journal - in italics*, Volume number (Issue number – in brackets), followed by page numbers.

Two authors:

Bibliography example:

Trefts, K., & Blaksee, S. (2000). Application of Image Processing For development of Automated Inspection System. *International Journal of Computational Engineering Research*, 28 (4), 369-378.

In-text examples:

(Trefts & Blaksee, 2000)

This supports Trefts and Blaksee’s (2000) evidence that...

“Direct quotations are placed in double quotations marks” (Author’s Surname, Year of Publication, p. – followed by page number – in brackets)

Journal article (online/electronic journal article)

Online article format:

Family/Surname, Initials. (Year of publication – in brackets). Title of article. Title of journal - in italics., Volume number (Issue number – in brackets), followed by page numbers. doi: xxxxxxxx

Single author:

Bibliography example:

Wilson, J. (1995). Enter the cyberpunk librarian: future directions in cyberspace. *Library Review*, 44(8), 63-72. doi: <http://dx.doi.org/10.1103/08670721311350>

In-text examples:

(Wilson, 1995)

According to Wilson (1995)....

“Direct quotations are placed in double quotations marks” (Author’s Surname, Year of Publication, p. – followed by page number – in brackets)

Conference Papers

Format:

Family/Surname, Initials. (Year of publication - in brackets). Title of Paper. *Full Title of Conference* - in italics, page numbers of paper.

Two authors:

Bibliography example:

Aryal, G., & Camsarapalli, S. K. (2010). Effect of Variables and their Interactions on RFID Tag Readability on a Conveyor Belt –Factorial Analysis Approach. *IEEE International Conference on Electro/Information Technology*, 1-6.

In-text example:

(Aryal & Camsarapalli, 2010)

Aryal & Camsarapalli (2010) noted that.....

“Direct quotations are placed in double quotations marks” (First Author’s Surname & Second Author’s Surname, Year of Publication, p. – followed by page number – in brackets)

Three or more authors:

Bibliography example:

Aryal, G., Mapa, L., & Camsarapalli, S. K. (2010). Effect of Variables and their Interactions on RFID Tag Readability on a Conveyor Belt –Factorial Analysis Approach. *IEEE International Conference on Electro/Information Technology*, 1-6.

In-text example:

(Aryal et al., 2010)

Aryal et al. (2010) noted that.....

“Direct quotations are placed in double quotations marks” (First Author’s Surname et al., Year of Publication, p. – followed by page number – in brackets)

Appendix C

Definition of Complex Problem Solving:

The range of complex problem solving is defined as follows:

SN	Attribute	Complex Problems have characteristic WP1 and some or all of WP2 to WP7
WP1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamental-based, first principles analytical approach.
WP2	Range of Conflicting requirements	Involve wide-ranging and/or conflicting technical, nontechnical issues (such as ethical, sustainability, legal, political, economic,
WP3	Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.
WP4	Familiarity of issues	Involve infrequently encountered issues or novel problems.
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice for professional engineering
WP6	Extent of stakeholder involvement and level of conflicting requirements	Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
WP7	Interdependence	Address high level problems with many components or sub-problems that may require a systems approach

(Source: EAC STANDARD 2024)

Definition of Complex Engineering Activities:

The range of complex engineering activities is defined as follows:

SN	Attribute	Complex Activities mean (engineering) activities or projects that have some or all of the following characteristics:
EA1	Range of resources	Involve the use of diverse resources including people, data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
EA2	Level of Interactions	Require optimal resolution of interactions between wide-ranging and/or conflicting technical, nontechnical, and engineering issues
EA3	Innovation	Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge.
EA4	Consequences to society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.
EA5	Familiarity	Can extend beyond previous experiences by applying principles

(Source: EAC MAUAL 2024)

APPENDIX D

PEER REVIEW FORM GROUP DESIN PROJECT

PEER REVIEW FORM (To be filled in by each team member individually based on colleagues' contribution to project independently.)

*Submit this form individually to the lecturer before Week 14 or poster presentation sessional information is confidential.

Submit this form individually to the lecturer before week 11 or poster presentation sessional information is confidential.						
Peer Review (10%) (Individual)	0-1	2	3	4	5	Marks
Team member 1:	Demonstrated the Individual did not contribute to the project and failed to meet responsibilities and the Individual did not identify the key performance criteria of successful teams or draw inference to own experience and the demonstration was without any effort by the individual and the system developed was not working.	Demonstrated the Individuality with low contribution to the project but failed to meet all the responsibilities and the Individual did not identify the key performance criteria of successful teams or draw inference to own experience and the demonstration showed the initial effort made by the individual to show the partially working system developed.	Demonstrated the Individuality with an average contribution to the project but did meet all the responsibilities and the Individual is also able to identify some of the key performance criteria of successful teams and / or draw related connections for the group performance and the demonstration showed the minimal effort made by the individual to show the fully working system developed.	Demonstrated the Individuality but did not contribute as heavily as others still met all the responsibilities and the Individual is also able to identify some of the key performance criteria of successful teams and / or draw related connections for the group performance and the demonstration showed the average effort made by the individual to show the fully working system developed.	Demonstrated the individual contribution in a valuable way to the project and the Individual is also able to articulate the key performance criteria of successful teams and evaluate the group performance accordingly and the demonstration showed the strong effort made by the individual to show the fully working system developed.	
Team member 2:						
Team member 3:						
Team member 4:						
Team member 5:						
Comments						

Signature:

Name / TP:

You can only choose a single mark point per person (i.e If team member 1 gets 5, Mark point 5 is taken out of the mark pool and only 0 - 4 remains, and it repeats until the last member is assess. Each mark point can only be assigned to one member, and the highest is 5 to the lowest 1, assigned based on degree of contribution to the project among peers. Mark point '0' is for members who did not contribute anything to the project. In the event of a tie, the evaluators will weigh in.

Total Mark = (Sum of all marks/number of members assessed)

APPENDIX E

Group Design Project Proposal Presentation Form

Module Name & Code:	
Group Project Title:	
Group Leader Name & TP Number:	
Group Member Names & TP numbers:	
Description	Marks
Group Project Background	
	3
Clearly explained the existing situation and problem related to the project.	3,2
Lack of explanation related to the project.	1
Failed to provide and explain the existing system.	0
Problem Statement	
	3
Clearly stated the current or intended problem to be solved.	3,2
Problem to solve is stated but lacks on explanation.	1
Failed to state the current or intended problem to be solved.	0
Aim and Objectives	
	3
Clear aim and objectives to the problem stated.	3,2
Clear aim and objectives but minimally related to the problem stated.	1
Aim and objectives are not related to the problem stated.	0
Justification for this project	
	2
Justification provided is clearly intended to solve the problem stated in this project.	2
Justification provided is intended to solve the problem stated but lacks on clear explanation.	1
Failed to justify the reasons for this project.	0
Literature Review	
	2
Review carried out is highly relevant to the project.	2
Review carried out is relevant to the project.	1
No review has been carried.	0
Proposed Methodology	
	10
Proposed methodology is able to solve the complex engineering problem stated with sufficient level of engineering solution based on the independent research done.	7,8,9,10
Proposed methodology is able to solve the complex engineering problem stated with fairly adequate level of engineering solution based on the independent research done.	4,5,6
Proposed methodology is able to solve the complex engineering problem stated with limited level of engineering solution based on the independent research done.	1,2,3
Failed to propose methodology to solve the complex engineering problem with no evidence of research done.	0
Innovativeness/Creativity/Enhancement	
	3
High level of new skills/techniques/creativity/enhancement have been proposed.	3,2
Fewer new skills/techniques/creativity/enhancement have been proposed.	1
No new skills/techniques/creativity/enhancement.	0
Sustainability	
	2
Well-developed demonstration of sustainability in this project.	2
Developed demonstration of sustainability in this project.	1
No evidence of sustainable development in this project.	0
Gantt Chart	
	2
Provided with feasible plan to complete the project within feasible timeline, with feasible work breakdown.	2
Provided with feasible plan to complete the project within feasible timeline, but work breakdown is not feasible.	1
Failed to provide a feasible plan to complete the project within feasible timeline nor feasible work breakdown.	0
Total Group Mark (30 Marks)	
CLO6-PLO8-Final Scaled Group Percentage (5%)	

Group Design Project Mid-Point Presentation Form

Module Name & Code:				
Group Project Title:				
Group Member Name & TP Number:				
A. B. C. D. E. F.				
Student	A	B	C	D
Description	Marks			
Individual Progress and Participation in Group Work [CLO6-PLO8]	5	5	5	5
Individual demonstrated a very good progressive design / simulation / partial prototype. Individual completed all assigned tasks before the deadline and had a level of engagement that demonstrated a strong commitment to the class and the learning outcome. Individual's accomplished work is thorough, comprehensive, and advances the project. Individual actively helps other team members to complete tasks to a similar level of excellence.	4,5	4,5	4,5	4,5
Individual demonstrated a good progressive design / simulation / partial prototype. Individual completes tasks by or just after the deadline and has a level of engagement that demonstrated a commitment to the class and / or the learning outcome. Individual offers some suggestions to advance the work of the group. Individual engages team members by restating the views of other team members and / or asking questions for clarification, if any.	2,3	2,3	2,3	2,3
Individual demonstrated a better progressive design / simulation / partial prototype. Individual shares idea but do not advance the work of the group and has a level of engagement that did not demonstrate a commitment to the class or the learning outcome.	1	1	1	1
Individual did not make any progress and did not contribute to the project and failed to meet the responsibilities.	0	0	0	0
Group Work [CLO6-PLO8]	5	5	5	5
The group or team worked well together to achieve objectives. Each member contributed in a valuable way to the project. Progress done indicates a high level of mutual respect and collaboration. Full progress made and is according to the Gantt chart and also, demonstrated the further plan with timeline to complete the group design project.	4,5	4,5	4,5	4,5
The group or team worked well together most of the time, with only a few occurrences of communication breakdown or failure to collaborate when necessary. Members were mostly respectful of each other. Partial progress made and is according to the Gantt chart and also, demonstrated the further plan with timeline to complete the group design project.	2,3	2,3	2,3	2,3
The group or team did not collaborate or communicate well. Some members would work independently, without regard to objectives or priorities. A lack of respect and regard was frequently noted. No progress made but demonstrated the further plan with timeline to complete the group design project.	1	1	1	1
No group or team work evident. No progress made and no further plan to complete the group design project.	0	0	0	0
CLO6-PLO8				
Total Individual Mark (5 Marks)				
Total Group Mark (5 Marks)				

Group Design Project Final Individual Presentation Form

Module Name & Code:				
Group Project Title:				
Group Member Name & TP Number:				
A. B. C. D.				
Student	A	B	C	D
Description	Marks			
Individual Proposed Design Solution and Participation in Group Work [CLO6-PLO8]	5	5	5	5
Individual demonstrated a very good design / simulation / prototype fully with sufficient level of engineering solution. Individual completed all assigned tasks before the deadline and had a level of engagement that demonstrated a strong commitment to the class and the learning outcome. Individual's accomplished work is thorough, comprehensive, and advances the project. Individual actively helps other team members to complete tasks to a similar level of excellence.	4,5	4,5	4,5	4,5
Individual demonstrated a good design / simulation / prototype with fairly adequate level of engineering solution. Individual completes tasks by or just after the deadline and has a level of engagement that demonstrated a commitment to the class and / or the learning outcome. Individual offers some suggestions to advance the work of the group. Individual engages team members by restating the views of other team members and / or asking questions for clarification, if any.	2,3	2,3	2,3	2,3
Individual demonstrated a better design / simulation / prototype with limited level of engineering solution. Individual shares idea but do not advance the work of the group and has a level of engagement that did not demonstrate a commitment to the class or the learning outcome.	1	1	1	1
Individual did not make any progress and did not contribute to the project and failed to meet the responsibilities.	0	0	0	0
Group Work [CLO6-PLO8]	5	5	5	5
The group or team worked well together to achieve objectives. Each member contributed in a valuable way to the project. Progress done indicates a high level of mutual respect and collaboration. Full progress made and is according to the Gantt chart and also, demonstrated the further plan with timeline to complete the group design project.	4,5	4,5	4,5	4,5
The group or team worked well together most of the time, with only a few occurrences of communication breakdown or failure to collaborate when necessary. Members were mostly respectful of each other. Partial progress made and is according to the Gantt chart and also, demonstrated the further plan with timeline to complete the group design project.	2,3	2,3	2,3	2,3
The group or team did not collaborate or communicate well. Some members would work independently, without regard to objectives or priorities. A lack of respect and regard was frequently noted. No progress made but demonstrated the further plan with timeline to complete the group design project.	1	1	1	1
No group or team work evident. No progress made and no further plan to complete the group design project.	0	0	0	0
CLO6-PLO8				
Total Individual Mark (5 Marks)				
Total Group Mark (5 Marks)				

Group Design Project Poster Presentation Form

Module Name & Code:	
Group Project Title:	
Group Member Names & TP Numbers:	
A. B. C. D.	
Description	Marks
Problem Statement	
3	
Clearly stated the current or intended problem to be solved.	3,2
Problem to solve is stated but lacks on explanation.	1
Failed to state the current or intended problem to be solved.	0
Aim and Objectives	
2	
Clear aim and objectives to the problem stated.	2
Clear aim and objectives but minimally related to the problem stated.	1
Aim and objectives are not related to the problem stated.	0
Justification for this project	
2	
Justification provided is clearly intended to solve the problem stated in this project.	2
Justification provided is intended to solve the problem stated but lacks on clear explanation.	1
Failed to justify the reasons for this project.	0
Proposed Methodology	
5	
Proposed methodology is able to solve the problem stated with sufficient level of engineering solution.	4,5
Proposed methodology is able to solve the problem stated with fairly adequate level of engineering solution.	2,3
Proposed methodology is able to solve the problem stated with limited level of engineering solution.	1
Failed to propose methodology to solve the problem stated.	0
Testing & Evaluation	
5	
System designed is tested with adequate data collection and is evaluated appropriately.	4,5
System designed is tested with adequate data collection but is not evaluated appropriately.	2,3
System designed is tested with data collection but not adequately and is not evaluated.	1
System designed is not tested.	0
Innovativeness/Creativity/Enhancement	
3	
High level of new skills/techniques/creativity/enhancement have been used.	3,2
Fewer new skills/techniques/creativity/enhancement have been used.	1
No new skills/techniques/creativity/enhancement.	0
Sustainability	
2	
Well-developed demonstration of sustainability in this project.	2
Developed demonstration of sustainability in this project.	1
No evidence of sustainable development in this project.	0
Marketability	
3	
Able to market easily and provide the actual cost incurred.	3,2
Able to market but without the actual cost incurred.	1
Not able to market.	0
Effective Oral Communication	
15	
Explains points explicitly with specific explanatory detail; points relate to overall argument. Presents information in an appropriate, logical order for the complex engineering activity taken. The style and tone of the communication enhances its effectiveness; the information has discernible style and elegant.	12,13,14,15
Most points are elaborated upon with specific explanatory detail; and most points relate to overall argument. Organizes information in a mostly logical pattern for the complex engineering activity taken. The style and tone of the communication supports its effectiveness.	8,9,10,11
Points need further elaboration, many points do not relate to overall argument, and meaning is often unclear.	4,5,6,7

Makes some attempt to order information for the complex engineering activity taken. The style and tone of the communication supports effectiveness in some aspects and undermines it in others.	
Points are vague, with no relation to overall argument, and meaning is unclear. Lacks understandable pattern of organization for the complex engineering activity taken. The style and tone of the communication undermines its effectiveness.	0,1,2,3
Total Group Mark (40 Marks)	
CLO7-PLO9-Final Scaled Group Percentage (10%)	

Self-Review Template

Personal Information

- **Name:**
- **Position:**
- **Department:**
- **Date:**

1. Collaboration

- **Rating (1-5):**
- **Comments:**
 - Describe how you have collaborated with team members.
 - Provide examples of how you have supported the team and contributed to team success.
 - Identify any areas where you can improve your collaboration skills.

2. Communication

- **Rating (1-5):**
- **Comments:**
 - Reflect on your communication with the team.
 - Provide examples of effective communication and any challenges faced.
 - Suggest ways to improve your communication skills.

3. Reliability

- **Rating (1-5):**
- **Comments:**
 - Assess your ability to meet deadlines and deliver high-quality work.
 - Provide examples of your reliability and any instances where you struggled.
 - Identify strategies to improve your reliability.

4. Problem-Solving

- **Rating (1-5):**
- **Comments:**
 - Evaluate your problem-solving skills.
 - Provide examples of problems you have identified and solved.

- Suggest ways to enhance your problem-solving abilities.

5. Adaptability

- **Rating (1-5):**
- **Comments:**
 - Reflect on your ability to adapt to changes and handle multiple tasks.
 - Provide examples of how you have adapted to new situations.
 - Identify areas where you can improve your adaptability.

6. Overall Reflection

- **Summary:**
 - Summarize your overall performance and key takeaways from this self-review.
 - Highlight your strengths and areas for improvement.
 - Set personal goals for the next review period.

Meeting Mode Requirement Template

Criteria	Face-to-Face Meetings	Remote Meetings	Distributed Meetings
Communication	Demonstrates clear and immediate communication, with non-verbal cues easily observed.	Explains ideas clearly but may lack non-verbal cues; relies on technology.	Reports mixed clarity; depends on technology and time zones.
Engagement	Expresses high engagement due to physical presence.	Expresses moderate engagement; can be affected by distractions.	Expresses varied engagement; time zone differences can impact participation.
Collaboration	Demonstrates easy collaboration with direct interaction.	Demonstrates good collaboration using tools like video conferencing.	Demonstrates challenging collaboration due to asynchronous communication.
Technology Dependence	Justifies low dependence; minimal technology required.	Justifies high dependence; relies on stable internet and conferencing tools.	Justifies very high dependence; requires robust technology and coordination.
Flexibility	Proposes limited flexibility; requires physical presence.	Proposes high flexibility; allows remote participation from anywhere.	Proposes very high flexibility; accommodates different locations and time zones.
Cost	Reports higher costs due to travel and venue expenses.	Reports lower costs; no travel required.	Reports mixed costs; depends on technology and coordination efforts.
Team Dynamics	Demonstrates strong team bonding and rapport.	Demonstrates moderate bonding; can feel isolated.	Demonstrates varied dynamics; rapport can be challenging.
Productivity	Expresses high productivity with focused environment.	Expresses moderate productivity; can be affected by home environment.	Expresses varied productivity; depends on coordination and time zone management.
Decision Making	Explains quick decision-making with immediate feedback.	Explains effective decision-making with good tools.	Explains slower decision-making due to asynchronous communication.
Accessibility	Justifies limited accessibility to those who can physically attend.	Justifies accessibility for anyone with internet access.	Justifies high accessibility; accommodates diverse locations.

APPENDIX F- MARKING RUBRICS

CLO1-PLO4-CRITICAL THINKING

Criteria	Fail	Marginal Fail	Pass	Credit	Distinction
	0	1	2	3	4-5
Identification of Issues <i>(Summarised)</i>	Summarised poorly, failing to identify issues or literature grounding.	Summarised issues weakly, with limited literature relevance.	Summarised some issues, with basic literature reference.	Summarised most key issues, supported by relevant literature.	Summarised all key issues clearly, fully grounded in highly relevant literature.
	0	1	2	3	4-5
Use of Framework <i>(Explained)</i>	Explained ineffectively, failing to apply framework or literature.	Explained the framework weakly, with poor literature connection.	Explained the framework with basic structure, limited literature integration.	Explained the framework with moderate effectiveness and partial literature support.	Explained the chosen framework effectively, integrating literature to support analysis.
	0	1	2	3	4-5
Evidence and Support <i>(Justified)</i>	Justified ineffectively, with little or no evidence or literature support.	Justified weakly with incomplete evidence, poor literature use.	Justified arguments with basic evidence, minimal literature support.	Justified arguments with good evidence and some literature integration.	Justified arguments with strong, well-integrated evidence from literature and other sources.
	0	1	2	3	4-5



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Analysis and Synthesis (Evaluated)	Evaluated ineffectively, with no coherent analysis or literature engagement.	Evaluated poorly, synthesis minimal, literature underused.	Evaluated basically, weak synthesis, limited literature use.	Evaluated reasonably, analysing information with some literature use.	Evaluated thoroughly, synthesising information from literature to draw well-reasoned conclusions.
Application of Knowledge (Explained)	0	1	2	3	4-5
	Explained ineffectively, with no literature-informed reasoning.	Explained application attempts, lacking depth and literature relevance.	Explained applications with limited effectiveness, weak literature connection.	Explained applications with some effectiveness, literature partially informed reasoning.	Explained and applied insights from literature and critical thinking effectively to real-world scenarios.
Reflection and Evaluation (Reviewed)	0	1	2	3	4-5
	Reviewed ineffectively, with no reflection or literature evaluation.	Reviewed superficially, literature evaluation absent.	Reviewed reflection basically, with limited evaluation of literature.	Reviewed the framework with some depth, acknowledging literature limitations.	Reviewed the framework deeply, appraising literature relevance and limitations.
			Total Mark (30 Marks)		
			CLO1-PLO4-Final Scaled Percentage (5%)		



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CLO6-PLO8-Individual Effectiveness, Teamwork & Professional Skills

Criteria	Fail	Marginal Fail	Pass	Credit	Distinction
	0	1-4	5-6	7-8	9-10
Collaboration (Demonstrate)	▪ Demonstrated no collaboration or support for team members.	▪ Demonstrated minimal collaboration, providing little support to team members.	▪ Demonstrated adequate collaboration, supporting team members occasionally.	▪ Demonstrated good collaboration, consistently supporting team members.	▪ Demonstrated outstanding collaboration, consistently supporting team members and significantly contributing to team success.
	0	1-4	5-6	7-8	9-10
Communication (Explain)	▪ Explained ideas poorly, causing frequent misunderstandings.	▪ Explained ideas weakly, with occasional misunderstandings.	▪ Explained ideas adequately, though with some gaps in clarity.	▪ Explained ideas well, with minor lapses in clarity.	▪ Explained ideas clearly, effectively, and consistently, ensuring strong team communication.
	0	1-4	5-6	7-8	9-10
Reliability (Report)	▪ Reported tasks late and delivered poor-quality work.	▪ Reported tasks inconsistently, frequently missing deadlines	▪ Reported tasks adequately, sometimes meeting deadlines with satisfactory work.	▪ Reported tasks reliably, usually meeting deadlines and delivering good-quality work.	▪ Reported tasks consistently, meeting deadlines and delivering high-quality work.



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		producing subpar work.			
Problem-Solving (Propose)	0	1-4	5-6	7-8	9-10
	Proposed no solutions and failed to solve problems.	Proposed weak solutions, struggling to solve problems.	Proposed basic solutions with some guidance, solving problems adequately.	Proposed good solutions, effectively solving problems.	Proposed innovative solutions proactively, solving problems with creativity and insight.
Adaptability (Justify)	0	1-4	5-6	7-8	9-10
	Justified no ability to adapt, failing to handle changes or tasks.	Justified weak adaptability, struggling with changes and tasks.	Justified adequate adaptability, handling changes with some difficulty.	Justified good adaptability, handling changes and tasks effectively.	Justified excellent adaptability, easily handling changes and multiple tasks efficiently.
			Total Mark (50 Marks)		
			CLO6-PLO8-Final Scaled Percentage (5%)		



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CLO6-PLO8-Meeting Modes

Criteria	Fail	Marginal Fail	Pass	Credit	Distinction
	0	1-4	5-6	7-8	9-10
Face-to-Face Meetings	Demonstrates poor collaboration despite physical presence, struggles to explain ideas clearly, does not express reliability, fails to propose solutions, and cannot justify adaptability in a traditional setting.	Demonstrates minimal collaboration, explains ideas with frequent gaps, expresses limited reliability, weakly proposes solutions, and struggles to justify adaptability in face-to-face settings.	Demonstrates adequate collaboration, explains ideas with some clarity, expresses reliability occasionally, proposes basic solutions with guidance, and justifies adaptability with some difficulty in face-to-face meetings.	Demonstrates strong collaboration, explains ideas well, expresses reliability by usually meeting expectations, proposes effective solutions, and justifies adaptability by adjusting well to changes in face-to-face meetings.	Demonstrates outstanding collaboration, explains ideas clearly and consistently, expresses full reliability, proposes innovative solutions proactively, and justifies adaptability by easily handling challenges in face-to-face meetings.
	0	1-4	5-6	7-8	9-10
Remote Meetings	Demonstrates weak collaboration via technology, struggles to explain ideas without non-verbal cues,	Demonstrates basic collaboration with tools, explains ideas with lapses, expresses limited reliability, weakly proposes solutions, and	Demonstrates adequate collaboration using conferencing tools, explains ideas with moderate clarity, expresses reliability occasionally,	Demonstrates strong collaboration via technology, explains ideas well despite fewer non-verbal cues, expresses reliability by usually meeting expectations, proposes	Demonstrates outstanding collaboration with remote tools, explains ideas clearly and consistently, expresses full reliability, proposes



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	inconsistently expresses reliability, rarely proposes workable solutions, and fails to justify adaptability in remote contexts.	struggles to justify adaptability remotely.	proposes basic solutions, and justifies adaptability with some difficulty in remote settings.	effective solutions, and justifies adaptability by adjusting well in remote contexts.	innovative solutions proactively, and justifies adaptability by easily handling remote challenges.
	0	1-4	5-6	7-8	9-10
Distributed Meetings	Demonstrates poor collaboration across time zones, struggles to explain ideas with clarity, does not express reliability in asynchronous work, fails to propose solutions, and cannot justify adaptability in distributed teams.	Demonstrates minimal collaboration asynchronously, explains ideas inconsistently, expresses limited reliability, weakly proposes solutions, and struggles to justify adaptability across time zones.	Demonstrates adequate collaboration asynchronously, explains ideas with some clarity, expresses reliability occasionally, proposes basic solutions, and justifies adaptability with some difficulty in distributed teams.	Demonstrates strong collaboration across time zones, explains ideas well with technology, expresses reliability by usually meeting expectations, proposes effective solutions, and justifies adaptability by adjusting well in distributed teams.	Demonstrates outstanding collaboration across diverse locations, explains ideas clearly and consistently, expresses full reliability, proposes innovative solutions proactively, and justifies adaptability by easily handling distributed team challenges.
			Total Mark (30 Marks)		
			CLO6-PLO8-Final Scaled Percentage (5%)		

CLO7-PLO9-Effective Presentation for Different Learning Styles

Criteria	Fail	Marginal Fail	Pass	Credit	Distinction
	0	1-4	5-6	7-8	9-10
Visual Learners (Display)	Poster displayed no visuals or visuals were irrelevant.	Poster displayed visuals weakly, with minimal relevance to the content.	Poster displayed some visuals that supported understanding.	Poster displayed visuals well, with clear images, charts, or diagrams that enhanced comprehension.	Poster displayed visuals outstandingly, with a rich variety of images, charts, and diagrams that strongly enhanced understanding and engagement.
	0	1-4	5-6	7-8	9-10
Auditory Learners (Defend)	Presenter defended content poorly, with no oral support or audio/video elements.	Presenter defended content weakly, with minimal oral support or audio/video elements.	Presenter defended content adequately, with some oral explanation or audio/video elements that supported understanding.	Presenter defended content well, with clear oral explanation and effective audio/video elements.	Presenter defended content outstandingly, with high-quality oral delivery and audio/video elements that reinforced and enriched the presentation.
	0	1-4	5-6	7-8	9-10
Kinesthetic Learners (Exemplify)	Poster exemplified no interactive elements, failing	Poster exemplified interactive elements weakly, with minimal engagement.	Poster exemplified interactive elements adequately, with some engagement	Poster exemplified interactive elements well, with clear activities or	Poster exemplified interactive elements outstandingly, with creative, hands-on activities or



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	to engage the audience.		(e.g., QR codes, small activities).	demonstrations that engaged the audience.	demonstrations that strongly engaged and involved the audience.
Social Learners (Display)	0	1-4	5-6	7-8	9-10
	Poster displayed no opportunities for group discussion or collaboration.	Poster displayed group activities weakly, with minimal collaboration.	Poster displayed group activities adequately, with some opportunities for collaboration.	Poster displayed group activities well, encouraging meaningful discussion and collaboration.	Poster displayed group activities outstandingly, fostering strong collaboration, dialogue, and peer learning.
Solo Learners (Defend)	0	1-4	5-6	7-8	9-10
	Poster defended no opportunities for individual reflection or self-paced learning.	Poster defended reflection opportunities weakly, with minimal relevance.	Poster defended reflection opportunities adequately, with some prompts or takeaway materials.	Poster defended reflection opportunities well, providing clear prompts and self-paced learning opportunities.	Poster defended reflection opportunities outstandingly, with strong prompts, takeaway materials, and self-guided learning pathways.
			Total Mark (50 Marks)		
			CLO7-PLO9-Final Scaled Percentage (10%)		

Final Consolidated Rubrics:

Criteria	Fail	Marginal Fail	Pass	Credit	Distinction
Project Proposal presentation (5%) (Group) CLO6–A3-PLO9	0 Justified very poor or not justified any proposal as a team.	1 Justified the proposed methodology based on the independent research done, but not according to the complex engineering problem stated as a team.	2 Justified the proposed methodology based on the independent research done, with limited level of engineering solution for the complex engineering problem stated as a team.	3 Justified the proposed methodology based on the independent research done, with adequate level of engineering solution for the complex engineering problem stated as a team.	4-5 Justified the proposed methodology based on the independent research done, with sufficient level of engineering solution for the complex engineering problem stated as a team.
Literature Review (5%) (Individual Chapter 2/3/4/5) CLO1-C5-PLO4	0 Reviewed methods/ design through literature review is very poor but irrelevant to the topic with outcomes/findings and drawbacks/limitations/gaps for future work, evidenced by a critical thinking framework s.	1 Reviewed methods/ design through literature review is poor but relevant to the topic with no outcomes/findings and drawbacks/limitations/gaps for future work, evidenced by a critical thinking framework s.	2 Reviewed methods/ design through literature review is average with outcomes/findings and drawbacks/limitations/gaps for future work, evidenced by a critical thinking framework s.	3 Reviewed methods/ design through literature review is good and highly relevant to the topic with outcomes/findings and drawbacks/limitations/gaps for future work, evidenced by a critical thinking framework s.	4-5 Reviewed methods/ design through literature review is excellent and highly relevant to the topic with outcomes/findings and drawbacks/limitations/gaps for future work, evidenced by a critical thinking framework s.
Electronic / Mechatronic Circuit Design and Development (10%) Applicable / Not Applicable (Individual Chapter 2/ 3/ 4/ 5) CLO2-C6-PLO3	0 Design circuits not shown and the final design badly sketched with no labeling/ did not explain on the working principle.	1-4 Designed very poor or did not show any circuit designs and sketches not clear and neat and not labeled well and the working principle is barely explained.	5-6 Designed but did not show all circuit designs and developed sketches are not clear and neat and not labeled well and the working principle is briefly explained.	7-8 Designed all circuits and developed all sketches but might not be clear and neat and not labeled well and the working principle is sufficiently explained.	9-10 Designed all circuits and developed all sketches clearly and neat and labeled well and the working principle is clearly explained.

Criteria	Fail	Marginal Fail	Pass	Credit	Distinction
	0	1-4	5-6	7-8	9-10
GUI Design Development (10%) Applicable / Not Applicable (Individual Chapter 2/3/4/5) CLO2-C6-PLO3	Designed GUI is very poor or designs not shown and not done.	Designed GUI is poor or did not show any GUI design, not clear and neat and not labeled well and explained barely the Pseudo codes of the developed program	Designed GUI shown is not clear and neat and not labeled well and explained briefly on Pseudo codes of the developed program	Designed GUI shown might not be clear and not labeled well and explained sufficiently on the Pseudo codes of the developed program	Designed GUI shown is clear and neat and labeled well and explained clearly on the Pseudo codes of the developed program
Transceiver Design Development (10%) Applicable / Not Applicable (Individual Chapter 2/ 3/ 4/ 5) CLO2-C6-PLO3	Designed very poor transceiver or designs not shown or not done.	Designed poor or did not show any transceiver designs and sketches not clear and neat and not labeled well and explain barely the working principle.	Designed but did not show all transceiver designs and developed sketches are not clear and neat and not labeled well and explained briefly on the working principle.	Designed all transceivers and developed all sketches but might not be clear and neat and not labeled well and explained sufficiently on the working principle.	Designed all transceivers and developed all sketches clearly and neat and labeled well and explained clearly on the working principle.
Integrated System and Enhancement (10%) (Group) CLO2-C6-PLO3	- Proposed no modification. - Developed group circuit simulation and circuit proto type is not working. - Designed Bread-board is not working. - Developed overall prototype as a solution for the complex engineering problem was not working, by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and	- Proposed modification done is simple. - Developed group circuit simulation and circuit proto type is working partially. - Designed Bread-board is not fully working. - Developed the overall prototype as a solution for the complex engineering problem was not working, by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural,	- Proposed modification done is moderate. - Developed group circuit simulation is working properly and circuit proto type is working partially. - Design based on bread-board is working. - Developed the overall prototype as a solution for the complex engineering problem was working partially, by incorporating public health and safety, whole-life cost, net zero carbon as well as resource,	- Proposed modification done is unique in terms of the integrating existing techniques or systems. - Developed group circuit simulation is working properly and circuit proto type is working partially. - Designed PCB is not fully working. - Developed the overall prototype as a solution for the complex engineering problem was working, by incorporating public health and safety, whole-life cost,	- Proposed modification done is creative and/or innovative with new skills involved. - Developed group circuit simulation and circuit proto type is working properly. - Designed PCB is fully working. - Developed the overall prototype as a solution for the complex engineering problem was working, by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal,



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	environmental considerations.	societal, and environmental considerations.	cultural, societal, and environmental considerations.	net zero carbon as well as resource, cultural, societal, and environmental considerations.	and environmental considerations.
	0	1	2	3	4-5
Professional Engineering Practices (5%) (Group) CLO3-C5-PLO6	Explained professional engineering practices that were not relevant, with no consideration of safety, health, social, cultural, or legal aspects, nor any incorporation of social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, or UN-SDGs.	Explained professional engineering practices but did not address safety, health, social, cultural, or legal aspects, and failed to incorporate social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, or UN-SDGs.	Explained professional engineering practices for safety, health, social, cultural, and legal responsibilities in developing solutions for complex engineering problems, but some of the explanation was irrelevant. Limited incorporation of social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and fewer than ten UN-SDGs.	Explained professional engineering practices for safety, health, social, cultural, and legal responsibilities in developing solutions for complex engineering problems, with relevant coverage. Incorporated social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and addressed at least ten UN-SDGs, though with moderate depth.	Explained in detail the professional engineering practices for safety, health, social, cultural, and legal responsibilities in developing solutions for complex engineering problems. Fully incorporated social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and demonstrated integration of at least ten UN-SDGs with strong relevance and depth
Sustainability and Environmental considerations (10%) (Group) CLO4-C5-PLO7	0 Explained sustainability of design and environmental considerations were not addressed. No incorporation of social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, or UN-SDGs	1-4 Explained sustainability and environmental considerations linked to the proposed solution weakly. Minimal or no incorporation of social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and fewer than ten UN-SDGs.	5-6 Explained sustainability and environmental considerations linked to the design aspect adequately. Incorporated some elements of social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, but fewer than ten UN-SDGs or with limited relevance.	7-8 Explained sustainability and environmental considerations linked to the design aspect very well. Incorporated social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and addressed at least ten UN-SDGs with moderate depth and relevance.	9-10 Explained sustainability and environmental considerations linked to the design aspect outstandingly. Fully incorporated social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and demonstrated integration of at least ten UN-SDGs with strong relevance, depth,



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Project Management, Finance & Entrepreneurship (10%) (Group) CLO5-C5-PLO11	0 Explained project management, cost analysis and entrepreneurship is very poor or not given.	1-4 Explained project management with the help of Gantt chart, cost analysis and marketing strategy provided is poor.	5-6 Explained project management with the help of Gantt chart, and cost analysis is average and minimal identification of a target market provided.	7-8 Explained project management with the help of Gantt chart and cost analysis is brief and clear understanding of target market provided.	9-10 Explained project management with the help of Gantt chart and cost analysis is detailed, and excellent identification of one or more target markets with trends identified.
Mid-point Group presentation – Group Progress updates (5%) (Group) CLO6-A3-PLO8	0 - Demonstrated very poor or no group or team work evident. - Justified very poor or no progress made and no further plan to complete the group design project.	1 - Demonstrated the group or team did not collaborate or communicate well, some members would work independently, without regard to objectives or priorities and a lack of respect and regard was frequently noted. - Justified very poor or no progress made but demonstrated the further plan with timeline to complete the group design project.	2 - Demonstrated the group or team collaboration and communicated well only for a very few times, some members would work independently, without regard to objectives or priorities and members were almost respectful of each other - Justified with initial progress made but demonstrated the further plan with timeline to complete the group design project.	3 - Demonstrated the group or team worked well together most of the time, with only a few occurrences of communication breakdown or failure to collaborate when necessary and members were mostly respectful of each other - Justified partial progress made and is according to the Gantt chart and also, demonstrated the further plan with timeline to complete	4-5 - Demonstrated the group or team work well together to achieve objectives - Justified progress done indicates a high level of mutual respect and collaboration. - Justified full progress made and is according to the Gantt chart and also, demonstrated the further plan with timeline to complete the group design project.

Mid-point Group Presentation - Individual Progress updates (5%) (Individual) CLO6-A3-PLO8	0	1	2	3	4-5
	▪ Demonstrated the Individual did not contribute to the project and failed to meet responsibilities, and the Individual did not identify the key performance criteria of successful teams or draw inference to own experience and no progress and timelines are not meet.	▪ Demonstrated the Individuality with low contribution to the project but failed to meet all the responsibilities and the Individual did not identify the key performance criteria of successful teams or draw inference to own experience and poor progress and timelines are not meet.	▪ Demonstrated the Individuality with an average contribution to the project but did meet all the responsibilities and the Individual is also able to identify some of the key performance criteria of successful teams and / or draw related connections for the group performance and average progress and timelines are not frequently met.	▪ Demonstrated the Individuality but did not contribute as heavily as others still met all the responsibilities and the Individual is also able to identify some of the key performance criteria of successful teams and / or draw related connections for the group performance and reasonable progress and timelines are frequently met.	▪ Demonstrated the Individual contribution in a valuable way to the project and the Individual is also able to articulate the key performance criteria of successful teams and evaluate the group performance accordingly and good progress and timelines are met.
Final Individual presentation – Group Progress updates (5%) (Group) CLO6-A3-PLO8	0	1	2	3	4-5
	▪ Demonstrated very poor or no group or team work evident. ▪ Justified very poor or no progress made and no further plan to complete the group design project.	▪ Demonstrated the group or team did not collaborate or communicate well, some members would work independently, without regard to objectives or priorities and a lack of respect and regard was frequently noted. ▪ Justified very poor or no progress made but demonstrated the further plan with timeline to complete the group design project.	▪ Demonstrated the group or team collaboration and communicated well only for a very few times, some members would work independently, without regard to objectives or priorities and members were almost respectful of each other ▪ Justified with initial progress made but demonstrated the further plan with timeline to complete the group design project.	▪ Demonstrated the group or team worked well together most of the time, with only a few occurrences of communication breakdown or failure to collaborate when necessary and members were mostly respectful of each other ▪ Justified partial progress made and is according to the Gantt chart and also, demonstrated the further plan with timeline to complete	▪ Demonstrated the group or team work well together to achieve objectives ▪ Justified progress done indicates a high level of mutual respect and collaboration. ▪ Justified full progress made and is according to the Gantt chart and also, demonstrated the further plan with timeline to complete the group design project.

	0	1	2	3	4-5
Final Individual Presentation (5%) (Individual) CLO6-A3-PLO8	▪ Demonstrated the Individual did not contribute to the project and failed to meet responsibilities and the Individual did not identify the key performance criteria of successful teams or draw inference to own experience and the demonstration was without any effort by the individual and the system developed was not working.	▪ Demonstrated the Individuality with low contribution to the project but failed to meet all the responsibilities and the Individual did not identify the key performance criteria of successful teams or draw inference to own experience and the demonstration showed the initial effort made by the individual to show the partially working system developed.	▪ Demonstrated the Individuality with an average contribution to the project but did meet all the responsibilities and the Individual is also able to identify some of the key performance criteria of successful teams and / or draw related connections for the group performance and the demonstration showed the minimal effort made by the individual to show the fully working system developed	▪ Demonstrated the Individuality but did not contribute as heavily as others still met all the responsibilities and the Individual is also able to identify some of the key performance criteria of successful teams and / or draw related connections for the group performance and the demonstration showed the average effort made by the individual to show the fully working system developed.	▪ Demonstrated the individual contribution in a valuable way to the project and the Individual is also able to articulate the key performance criteria of successful teams and evaluate the group performance accordingly and the demonstration showed the strong effort made by the individual to show the fully working system developed.
	0	1	2	3	4-5
Individual Effectiveness (5%) (Individual) CLO6-A3-PLO8	▪ Demonstrated poor collaboration, struggles to explain ideas, does not express reliability, cannot propose solutions, and fails to justify adaptability.	▪ Demonstrated minimal collaboration, explains poorly, expresses limited reliability, struggles to propose solutions, and struggles to justify adaptability.	▪ Demonstrated adequate collaboration, explains with some gaps, expresses reliability occasionally, proposes basic solutions with guidance, and justifies adaptability with some difficulty.	▪ Demonstrated good collaboration, explains ideas well, expresses reliability by usually meeting deadlines, proposes effective solutions, and justifies adaptability by adjusting well to changes.	▪ Demonstrated active collaboration, clearly explains ideas, consistently expresses reliability, proposes innovative solutions, and justifies adaptability by easily adjusting to changes and handling tasks efficiently.
	0	1	2	3	4-5
Peer Review (5%) (Individual) [CLO6-A3-PLO9]	▪ Demonstrated the Individual did not contribute to the project and failed to meet responsibilities and the Individual did not identify the key performance criteria of successful teams or draw inference to own experience and the demonstration was without	▪ Demonstrated the Individuality with low contribution to the project but failed to meet all the responsibilities and the Individual did not identify the key performance criteria of successful teams or draw inference to own experience and the demonstration showed the	▪ Demonstrated the Individuality with an average contribution to the project but did meet all the responsibilities and the Individual is also able to identify some of the key performance criteria of successful teams and / or draw related connections for the group performance and	▪ Demonstrated the Individuality but did not contribute as heavily as others still met all the responsibilities and the Individual is also able to identify some of the key performance criteria of successful teams and / or draw related connections for the group performance and	▪ Demonstrated the individual contribution in a valuable way to the project and the Individual is also able to articulate the key performance criteria of successful teams and evaluate the group performance accordingly and the demonstration showed

	any effort by the individual and the system developed was not working.	initial effort made by the individual to show the partially working system developed.	the demonstration showed the minimal effort made by the individual to show the fully working system developed	the demonstration showed the average effort made by the individual to show the fully working system developed.	the strong effort made by the individual to show the fully working system developed.
	0	1	2	3	4-5
Meeting Modes (5%) (Group) CLO6-A3-PLO8	▪ Demonstrated poor collaboration, struggled to explain ideas clearly, did not express reliability, failed to propose workable solutions, and could not justify adaptability across face-to-face, remote, or distributed meetings.	▪ Demonstrated minimal collaboration, explained ideas with frequent gaps, expressed limited reliability, weakly proposed solutions, and struggled to justify adaptability in different meeting formats.	▪ Demonstrated adequate collaboration, explained ideas with some clarity, expressed reliability occasionally, proposed basic solutions with guidance, and justified adaptability with some difficulty across meeting modes.	▪ Demonstrated strong collaboration, explained ideas well, expressed reliability by usually meeting expectations, proposed effective solutions, and justified adaptability by adjusting well to changes across face-to-face, remote, and distributed meetings.	▪ Demonstrated outstanding collaboration, explained ideas clearly and consistently, expressed full reliability, proposed innovative solutions proactively, and justified adaptability by easily handling challenges across all meeting formats
	0	1-4	5-6	7-8	9-10
Poster Presentation/ IEEE paper - Integrated System working & demonstration (10%) (Group) CLO7-A5-PLO10	▪ Exemplified to fail or may not have turned up for communication and design is not complete. Communication did not take into account cultural, language, or learning differences (Visual, Auditory, Kinesthetic, Social, Solo learners). No IEEE paper	▪ Exemplified to fail to fully communicate and understand the principles and elements of the complex engineering activity. ▪ Defended the design without creativity and originality, with developing quality and presented in an acceptable manner with 1-2 successful launches. Inadequate time usage and little evidence of successful audience analysis. Communication weakly considered cultural, language, or learning differences, with little evidence of adaptation for diverse learners. IEEE	▪ Exemplified the principles and elements of the complex engineering activity through communication. ▪ Defended the design with originality, medium quality and presented in a somewhat professional manner with 2-3 successful launches. Adheres to time limitations and reflects appropriate analysis of the majority of the audience. Communication partially incorporated cultural, language, and learning differences, but inconsistently addressed Visual, Auditory, Kinesthetic, Social, and Solo learners. IEEE paper has	▪ Exemplified the principles and elements of the complex engineering activity through communication. ▪ Defended the design with originality, high quality and presented in a professional manner with 3-4 successful launches. Adheres to time limitations and reflects unusually insightful audience analysis. Communication incorporated cultural, language, and learning differences effectively, addressing Visual, Auditory, Kinesthetic, Social, and Solo learners. IEEE paper has minor	▪ Exemplified the principles and elements of the complex engineering activity through communication. ▪ Defended the design with creativity, high quality and presented in a professional manner with 4-5 successful launches. Adheres to time limitations and reflects unusually insightful audience analysis. Communication was outstanding, fully incorporating cultural, language, and learning differences, and effectively engaging Visual, Auditory, Kinesthetic, Social, and Solo learners both orally and in written context. Excellent



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		paper has significant formatting, Turnitin (more than 30%) and content Issues	some formatting, Turnitin (more than 15%) and content issues	formatting and content issues	IEEE paper with factual centric and well developed content.
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Group Component Mark = _____

$$\text{Work Contribution Index} = \frac{\text{Obtained Peer Evaluation Mark}}{\text{The Highest Group Member's Mark}}$$

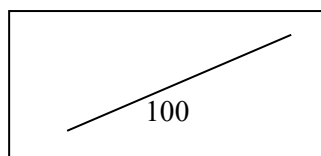
= _____ = _____

Work Contribution Index	Group Mark Index
0.95 – 1.00	1.0
0.90 – 0.94	0.9
0.85 – 0.89	0.7
0.75 – 0.84	0.6
0.55 – 0.74	0.5
0.35 – 0.54	0.3
0.25 – 0.34	0.1
0 – 0.24	0.0

Final Group Mark = Group Mark Index X Group Component Mark
=

Total Mark = Final Group Component Mark + Individual Component Mark
=

Total Marks



Lecturer: _____